Neoantimon2 — variants to traceable candidates

Native and independent External routes • Examples illustrative; references and predictors obtained separately

Patients + samples
P1 → S1, S2
One sample column / VCF

Reference bundle
GRCh38.p14 · GENCODE 50
MANE Select 1.5

Patient-level HLA
HLA-A*02:01 / HLA-B*07:02
Declared once per patient

Optional local predictors
Class I: NetMHCpan / MHCflurry
Class II: NetMHCIIpan

Optional observations
RNA / CCF / MS

A NATIVE

Variant events
chr1:4 G>A
Normalized variant

→ **Origins + translation**
Transcript TA.1
Translation path

→ **Peptide sequences**
SIINFEKLA
All origins retained

→ **Peptide–HLA candidates**
SIINFEKLA + HLA-A*02:01

→ **Prediction + evidence**
e.g. NOT_RUN
No score ≠ negative

→ **Prioritize + explain**
Rank / eligible grade
Candidate → origins

Multiple paths can share a peptide; every candidate–origin link is retained.

Completed Neoantimon2 project — scores / grades not required

B EXTERNAL — a separate derived view

Export bundle
Queries + all origins
Completed project

→ **Model outside Neoantimon2**
Run separately
User-selected model

→ **Standard format**
BA nM / EL rank / IM
Separate metric identity

→ **Strict matching**
Peptide or peptide + HLA

→ **External results + explanations**
Independent ranks
Origins + event trace

Import requires the original completed project. Strict matching preserves all origin links.
External grades require MHC percentiles + eligible core somatic evidence; raw scores alone do not grade.

Specialized workflows — explicit opt-ins; external scores do not feed these views

Longitudinal
Sample graphs + explicit settings
Native grades / ranks retained

Vaccine-set selection
Native evidence + explicit policy
Native grades / ranks retained

Restricted indirect-HLA / allo
Own inputs + restricted profile
Not an official PIRCHE implementation

Missing evidence ≠ zero ≠ evaluated negative • Research outputs, not clinical validation